Molecular Dynamics Simulations of Molten Magnesium Chloride Using Machine-Learning-Based Deep Potential

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In previous work, molten magnesium chloride has been investigated using first-principles molecular dynamics (FPMD) simulations based on density functional theory (DFT). However, such simulations are computationally intensive and therefore are restricted in terms of simulated size and time. In this work, a machine learning-based deep potential (DP) is trained to accelerate the molecular dynamics simulation of molten magnesium chloride. The trained DP can accurately describe the energies and forces with the prediction errors in energy and force being 1.76×10^{-3} eV/atom and 4.76×10^{-2} eV Å^{-1}, respectively. Applying the deep potential molecular dynamics (DPMD) approach, simulations can be performed with more than 1000 atoms, which is infeasible for FPMD simulations. Additionally, the partial radial distribution functions, angle distribution functions, densities, and self-diffusion coefficients predicted by DPMD simulations are also in reasonable agreement with FPMD or experimental results. This work shows that the DP enables higher efficiency and similar accuracy relative to DFT, exhibiting a bright application prospect in modeling molten salt systems.

1. Introduction

$MgCl_2$-based chloride salts have gained increasing attention as a promising candidate for heat transfer fluid (HTF) in the concentrated solar thermal power (CSP) systems.[1–4] Compared with the conventional nitrate/nitrite and carbonate salts, $MgCl_2$-based chloride salts have much higher decomposition temperature and lower material cost, which is beneficial to increase the energy conversion efficiency and decrease the cost of the CSP system.[5] It has been demonstrated that $MgCl_2$-KCl[1] and $MgCl_2$-$CaCl_2$-NaCl[2] eutectic chloride salts are satisfactory and acceptable to serve as a high-temperature HTFs. Understanding the macroscopic properties and microstructure of such salts will be helpful for the design of efficient HTFs. Unfortunately, experimental measurements on $MgCl_2$-based chloride salts are immensely difficult because of the high-temperature, corrosion, and evaporation involved.

The first-principles molecular dynamics (FPMD) method is the general approach to simulate $MgCl_2$-based chloride salts, in which the impact of polarization effect[6,7] on the microstructure of $MgCl_2$-based chloride salts can be fully considered with minimal human intervention. In earlier work, we have studied pure $MgCl_2$[8] and $MgCl_2$-KCl[9] binary salt through FPMD simulations. Similar first principle studies have been performed on $MgCl_2$-NaCl-KCl by Li et al.[3] and $MgCl_2$-$CaCl_2$-KCl by Rong et al.[4] to investigate the thermophysical properties and microstructural evolution. Despite the success of FPMD simulations, it has limitations; for example, large-scale and long-time molecular dynamics simulations are not accessible in the framework of the FPMD method.[10] An effort was made by Wu et al.[11] to address this issue, who developed an interatomic potential for $MgCl_2$-KCl binary salt base on the polarizable ion model (PIM).[12] However, fitting parameters of the PIM are usually a daunting task that is very time consuming and requires tremendous expertise. Additionally, human intervention is sometimes required to obtain reasonable results.

To generate interatomic potentials by machine learning (ML) approach is another way to solve this problem, which is more reliable and achievable. Using datasets extracted from FPMD calculations, the carefully trained ML potentials enable higher efficiency and similar accuracy relative to density functional theory (DFT). So far, several kinds of ML potentials have been reported in the literature, such as Behler–Parrinello neural network potential,[13] Gaussian approximation potential,[14] and the deep potential (DP).[15,16] In ML potentials training, the atomic coordinates cannot be used directly because their format does not preserve the symmetry of the system. Usually, the symmetry of the system is preserved by converting the atomic coordinates into vectors using a descriptor. The DP is adopted in this work since the descriptor proposed by Zhang et al.[15,16] for DP is considered to be more flexible. To our knowledge, the DP has been successfully applied in the investigations of various systems such as water,[17,18] ($Zr_{0.2}Hf_{0.2}Ti_{0.2}Nb_{0.2}Ta_{0.2}$)C high entropy materials,[19] solid-state electrolytes,[20] alloy materials,[21–23] etc.

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In this paper, a DP is trained for molten $MgCl_2$ using the DeePMD-kit package.[24] The accuracy of the trained DP is discussed by comparing energies and forces calculated with DP and DFT. Then, the local structure and transport properties obtained by deep potential molecular dynamics (DPMD) simulations are compared with those obtained by FPMD simulations and experimental measurements.

2. Experimental Section

2.1. DP Training

In this work, a scheme based on machine learning and deep neural networks (DNN) was employed to train deep potential for molten $MgCl_2$ using the DeePMD-kit package[24] (version 1.2.0). In the DeepMD method, the energy of a system is decomposed into a sum of atomic energy contributions, which ensures that the trained DP is workable for large-scale systems

$$E = \sum_i E_i \tag{1}$$

where E_i is determined by the local environment of atom i within a radius cutoff r_c

$$E_i = E_{s(i)} \left(\boldsymbol{R}_i, \{R_j | j \in N_{rc}(i)\} \right) \tag{2}$$

where $N_{rc}(i)$ denotes the index set of the neighbors of atom i. s(i) is the chemical species of atom i. During the training process, the atomic coordinates cannot be used directly because they are unable. To solve this problem, the coordinates of atoms R^i were converted to another format D^i in three steps. First, R^i was transformed into the generalized local environment matrix $\tilde{R}^i$

$$R^i = \{x_{ji}, y_{ji}, z_{ji}\} \to \tilde{R}^i = \{S(r_{ji}), \hat{x}_{ji}, \hat{y}_{ji}, \hat{z}_{ji}\} \tag{3}$$

where $\hat{x}_{ji} = \frac{S(r_{ji})x_{ji}}{r_{ji}}$, $\hat{y}_{ji} = \frac{S(r_{ji})y_{ji}}{r_{ji}}$, and $\hat{z}_{ji} = \frac{S(r_{ji})z_{ji}}{r_{ji}}$ are the angular information, and $S(r_{ji})$ is radial information

$$S(r_{ji}) = \begin{cases} \frac{1}{r_{ji}}, \ r_{ji} < r_{cs} \\ \frac{1}{r_{ji}} \left\{ \frac{1}{2}\cos\left[\pi \frac{r_{ji}-r_{cs}}{r_c - r_{cs}}\right] + \frac{1}{2} \right\}, r_{cs} < r_{ji} < r_c \\ 0, r_{ji} > r_c \end{cases} \tag{4}$$

where r_{cs} is smooth cutoff parameter. Second, the radial part of $\tilde{R}^i$ was mapped to the embedding matrix g^{i1} and g^{i2} by using an embedding neural network. Third, feature matrix D^i is defined through the product of matrices

$$D^i = (g^{i1})^T \tilde{R}^i (\tilde{R}^i)^T g^{i2} \tag{5}$$

The reader is encouraged to read the papers by Zhang et al.[15,16,24] for details. Then, the parameters of DNN were optimized through the minimization of the loss function

$$L(p_\epsilon, p_f, p_\xi) = p_\epsilon \Delta\epsilon^2 + \frac{p_f}{3N} \sum_i |\Delta F_i|^2 \tag{6}$$

where $\Delta\epsilon$ and ΔF_i are the energy difference per atom and force difference on atom i; N is the number of atoms; p_ϵ and p_f are tunable prefactors.

Datasets for DP training and validation were extracted from FPMD simulations of the previous work.[8] Here, an abbreviated description of the FPMD simulations is provided. All simulations were conducted by using the Vienna Ab Initio Simulation Package (VASP)[25,26] with projector-augmented wave (PAW)[27,28] for the interaction between electrons and ions and Perdew–Burke–Ernzerhof (PBE)[29] functional for the exchange-correlation functionals. The dispersion correction was taken into account by using the DFT-D2 method.[30,31] Besides, the wave functions for valence electrons, Mg 3s2 and Cl 3s23p5, were expanded using the plane-wave basis set with cutoff energy at 600 eV. The k-point mesh and time step were set to be $1 \times 1 \times 1$ and 1 ps for all calculations. The initial configuration was prepared using Packmol code,[32] which contained 36 Mg^{2+} cations and 72 Cl^- anions. The simulation cell was initially equilibrated at 2000 K for 6 ps and was subsequently quenched to multiple temperatures (1400,1200, and 1000 K) using a quench rate of 180 K ps^{-1}. Then, the equilibrium volume of the simulation cell corresponding to zero pressure at each temperature was obtained following the strategy of Bengtson et al.[33] Finally, simulations were run for 20 000 steps at each temperature. For each temperature, the first 75% was used for DP training and the last 25% was used for DP validation. Therefore, 60 000 snapshots were collected, in which 45 000 snapshots were collected for DP training and 15 000 snapshots were collected for DP validation. The embedding neural network has three hidden layers with 20, 50, and 100 nodes, respectively. The fitting neural network has three hidden layers with 240, 240, and 240 nodes, respectively. The smooth cutoff (r_{cs}) and the radius cutoff (r_c) were chosen to be 6.8 Å and 7 Å, respectively. The start values of tunable prefactors were 0.02 and 1000 for energy error and force error in the loss function, which means that the training process was dominated by force-fitting at the initial stage. At the end stage of the training process, the value of all tunable prefactors is equal to 1.

2.2. DPMD Simulations

All of the DPMD simulations were conducted using the Large-Scale Atomic/Molecular Massively Parallel Simulator (LAMMPS).[34] Molten $MgCl_2$ was investigated within the temperature range of 1000–1400 K with an interval of 100 K. So, five cubic simulation cells were prepared with randomly placed Mg (340 atoms) and Cl (680 atoms) using Packmol code.[32] For each temperature, the initial length (L) of the simulation cell was determined according to the corresponding experimental density[35] at the studied temperature. For all simulations, the periodic boundary condition was employed to eliminate the boundary effects. Newton's equation of motion was solved with a timestep of 1 fs using the Verlet algorithm.

The simulation flow was as follows: i) Assigned the initial velocities of the ions randomly at giving temperature according to a Gaussian distribution; ii) Equilibrated the system within the NPT ensemble at zero pressure 0 for 1000 ps; iii) The volume of each simulation cell was adjusted to the average between 600 and 1000 ps of simulations; iv) The production runs were per-

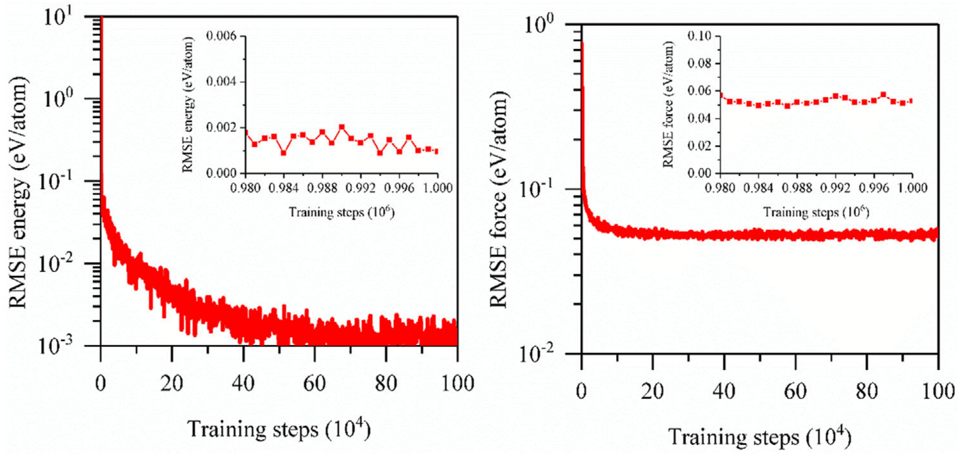

Figure 1. Evolution of energy (left) and force (right) RMSEs during the training process. The inset shows the RMSEs near the end of the training.

formed in *NVT* ensemble for 1000 or 3000 ps using the Nosé–Hoover thermostat. As in earlier studies,[36–38] the damping parameters were set to be 0.1 and 0.5 ps for thermostat and barostat, respectively.

2.3. Functions of Interest

2.3.1. Static Structure Information

The radial distribution functions (RDFs), also called pair correlation functions (PCFs), average coordination numbers (CN), and angular distribution functions (ADF) are frequently used to analyze the local structure of the ionic liquid

$$g_{\alpha\beta}(r) = \frac{1}{4\pi\rho_\beta r^2}\left[\frac{dN_{\alpha\beta}(r)}{dr}\right] \tag{7}$$

$$N_{\alpha\beta} = 4\pi r_\beta \int_0^{R_{\min}} g_{\alpha\beta}(r)\, r^2 dr \tag{8}$$

$$\theta_{jik} = \left\langle \cos^{-1}\frac{r_{ij}^2 + r_{ik}^2 - r_{jk}^2}{2r_{ij}r_{ik}} \right\rangle \tag{9}$$

where ρ_β is the number density of β ions and $N_{\alpha\beta}$ is the mean number of β-type ions lying in a sphere of radius r centered on an α-type ion. $R_{\min}$ is the position of the first minimum of the partial RDF.

2.3.2. Properties

The density was calculated according to the equilibrium volume of the *NPT* simulation through the following equation

$$\rho = \frac{\sum_i M_i N_i}{V_E N_A} \tag{10}$$

where M_i is the number of atom i, N_i is the molar mass of ion i, V_E is the equilibrium volume, and N_A is the Avogadro's constant.

The self-diffusion coefficients can be calculated from the mean square displacement (MSD) through the Einstein expression[39]

$$\mathrm{MSD} = \left\langle \left|\delta r_i(t)\right|^2 \right\rangle = \frac{1}{N}\left\langle \sum_i \left|r_i(t) - r_i(0)\right|^2 \right\rangle \tag{11}$$

$$D = \frac{1}{6}\lim_{t\to\infty}\frac{d(\mathrm{MSD})}{dt} \tag{12}$$

where $\delta r_i(t)$ is the displacement of a typical α-type ion in time t.

3. Results and Discussion

3.1. Accuracy of Trained DP

Figure 1 shows the root-mean-square errors (RMSEs) in energy and force evaluated from our DPs for training sets. It can be seen that the RMSE of energy and force decrease to ≈1.40 × 10^{-3} eV/atom and 5.25 × 10^{-2} eV Å^{-1}, respectively, after 1 × 10^6 training steps. Moreover, the trained DP is also tested using a dataset that is not used for the training process. The energies and forces predicted by the generated DP are plotted in **Figure 2**, and compared with the corresponding DFT results. RMSDs are 1.76 × 10^{-3} eV/atom and 4.76 × 10^{-2} eV Å^{-1} for energy and force, which are close to those in the training set. Since the energy RMSE of less than 2 × 10^{-3} eV/atom and the force RMSE of less than 6 × 10^{-2} eV Å^{-1} are sufficiently small, the trained DP is expected to be able to predict results with accuracy comparable to that of DFT.

3.2. Trajectories of DPMD Simulations

Simulations to generate DPMD simulation trajectories were run for 1000 ps in the *NVT* ensemble. RDFs and ADFs were obtained to further illustrate the faithfulness of these trajectories. The Mg—Cl, Mg—Mg, and Cl—Cl partial RDFs of molten $MgCl_2$ derived from DPMD and FPMD simulation trajectories are compared in **Figure 3**. As expected, the RDFs of DPMD simulations

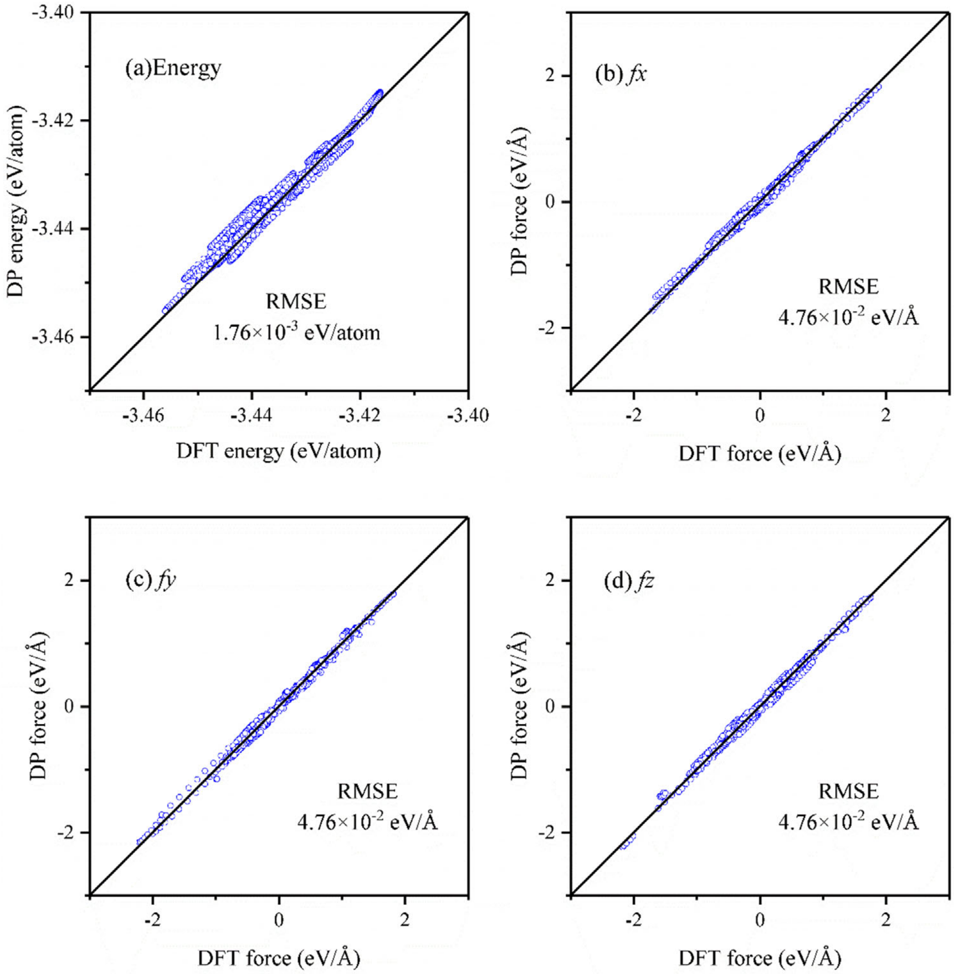

Figure 2. Comparison of a) energies and b–d) forces calculated with DP and DFT for a dataset that is not used for DP training.

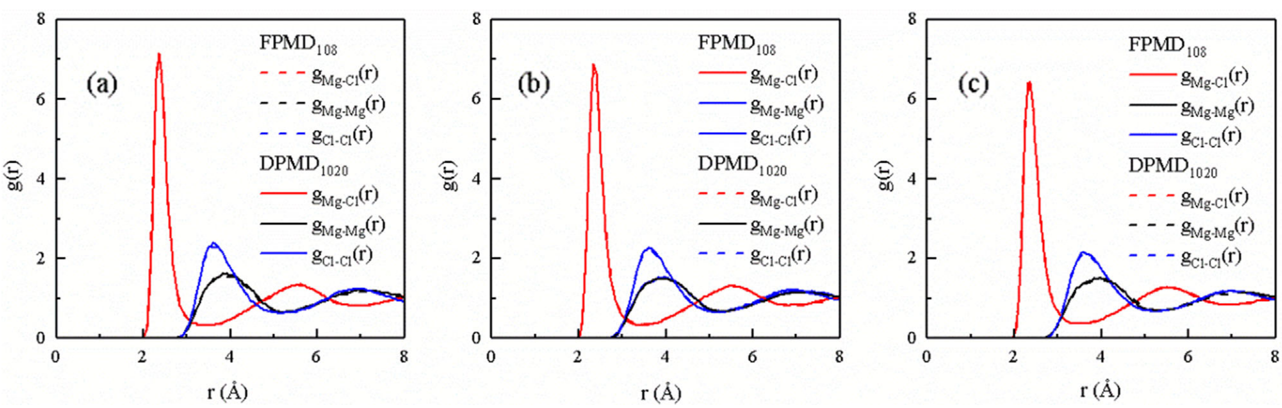

Figure 3. a–c) Radial distribution functions of molten $MgCl_2$ derived from DPMD and FPMD simulations at 1000, 1200, and 1400 K.

are in good agreement with those of DPMD simulations. In addition, the Cl-Mg-Cl ADFs derived from DPMD and FPMD simulation trajectories are compared in **Figure 4**. The Cl—Mg—Cl angles θ are calculated when two Cl^- anions are linked by a common Mg^{2+} cation and the corresponding Mg—Cl distances are shorter than the first minimum of Mg—Cl RDF. The results show that the trained DP acceptably reproduces the ADFs of FPMD simulations. A snapshot of molten $MgCl_2$ is given in **Figure 5**. As found in previous work,[8] the vast majority of Mg^{2+} cations are surrounded by four or five Cl^- anions. The trained DP does a reason-

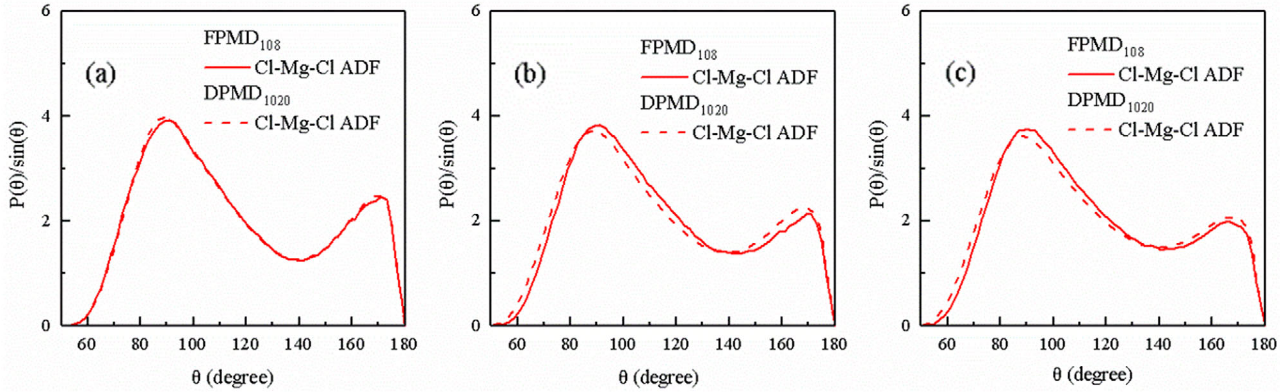

Figure 4. a–c) The Cl—Mg—Cl angular distribution functions derived from DPMD and FPMD simulations at 1000, 1200, and 1400 K. $P(\theta)/\sin\theta$ functions are shown; these distributions accurately represent the angular distribution functions while retaining sensitivity near 180°.[40]

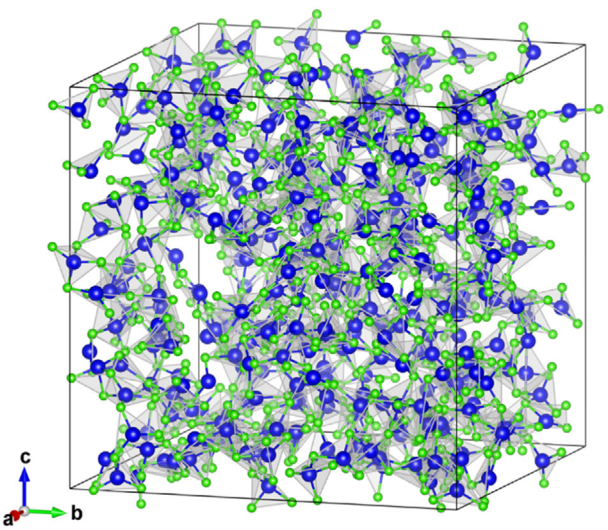

Figure 5. A snapshot of molten $MgCl_2$ extracted from the trajectory of DPMD simulations at 1000 K in the *NVT* ensemble. Mg^{2+} cations are marked in blue and Cl^- anions are marked in green.

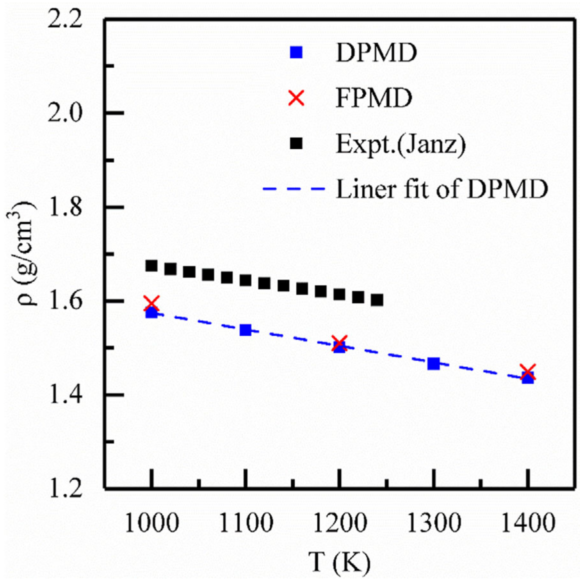

Figure 6. Densities of molten $MgCl_2$ evaluated from DPMD simulations as a function of temperature. Experimental data[35] and FPMD data[8] for molten $MgCl_2$ are shown together for comparison.

able job of reproducing the local structure of molten $MgCl_2$, although there are small differences. Since those differences were not significant, it is decided to calculate the basic properties of molten $MgCl_2$ using the DP.

3.3. Properties Predicted by DP

3.3.1. Density

All of the properties are calculated in the temperature range of 1000–1400 K. **Figure 6** shows the densities of molten $MgCl_2$ predicted by DPMD simulations as a function of temperature. Experimental densities for molten $MgCl_2$ were taken from Janz.[35] In addition, simulated densities evaluated from FPMD simulation are also included for comparison. It can be found that the simulated densities of DPMD are in close agreement with the FPMD results, but are slightly underestimated by ≈6% compared with the experimental values. The reason for the underestimated of simulated densities can be attributed to the inaccurate estimation of dispersion interaction by the commonly used density functional methods.[30] By a suitable choice of the dispersion correction methods, the difference between simulated and experimental densities can be reduced but not fully eliminated.[33] Moreover, the calculated density is negatively correlated to temperature and decrease from 1.57 g cm^{-3} at 1000 K to 1.502 g cm^{-3} at 1400 K, which can be fitted linearly as

$$\rho\left[\mathrm{gcm}^{-3}\right] = 1.924 - 3.50 \times 10^{-4}T\,[\mathrm{K}] \tag{13}$$

3.3.2. Self-Diffusion Coefficient

Self-diffusion coefficients of Mg^{2+} and Cl^- ions in pure $MgCl_2$ were calculated from the DPMD simulations according to the Einstein relationship between MSD and time.

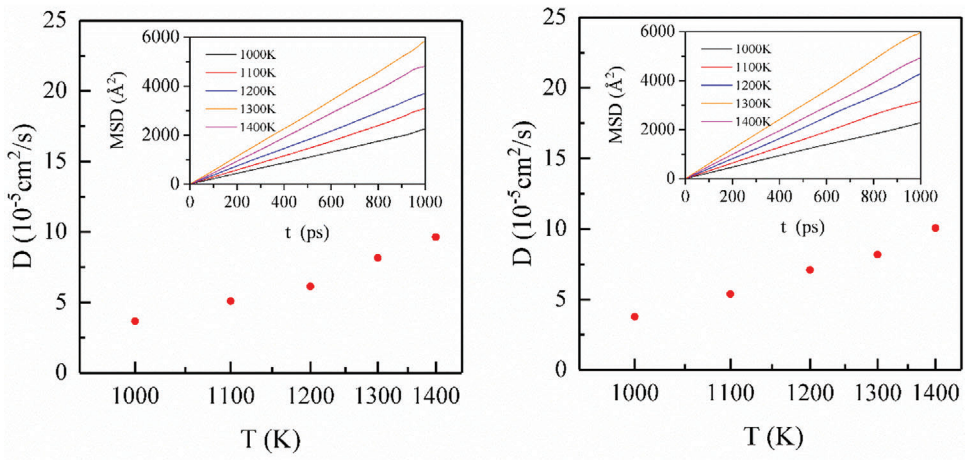

Figure 7. Temperature dependence of self-diffusion coefficients of Mg^{2+} (left) and Cl^- (right) ions in molten $MgCl_2$ from DPMD simulations. The inset shows MSDs of the ions as a function of elapsed time.

The estimated self-diffusion coefficients of ions in molten $MgCl_2$ at various temperatures are shown in **Figure 7**. To our knowledge, there is currently no experimental data available on self-diffusion coefficients of Mg^{2+} and Cl^- in pure $MgCl_2$, reflecting the difficulties of conducting experiments on molten $MgCl_2$ under high-temperature conditions. Although not plotted in Figure 7, FPMD results are lower than the values of DPMD simulations, which might be explained by the finite-size effect[41] (see Figures S1 and S2, Supporting Information). Additionally, FPMD simulations are usually conducted for short periods of time, which will reduce the accuracy and reliability of simulated results. The self-diffusion coefficients were fitted using an Arrhenius equation

$$\ln D_{Mg^{2+}} \left[10^{-4}\mathrm{cm^2 s^{-1}}\right] = -3370.28/T\,[\mathrm{K}] + 2.36723 \quad (14a)$$

$$\ln D_{Cl^-} \left[10^{-4}\mathrm{cm^2 s^{-1}}\right] = -3374.48/T\,[\mathrm{K}] + 2.42532 \quad (14b)$$

3.4. Computational Cost of DP

Here, we have summarized the computational efficiency of DPMD simulations for molten $MgCl_2$ with different system sizes, as shown in **Figure 8**. It reveals that the computational time of DPMD simulation scales linearly with the number of atoms in the system. The time required for 1 ns FPMD calculations with 108 atoms is about 7200 h. Therefore, DPMD simulation has a significant time advantage over DFT-based FPMD simulation which scales with the cube of the number of atoms. DPMD simulations become much more powerful tools when studying some physical properties that can only be calculated accurately in a larger system or longer timescale.

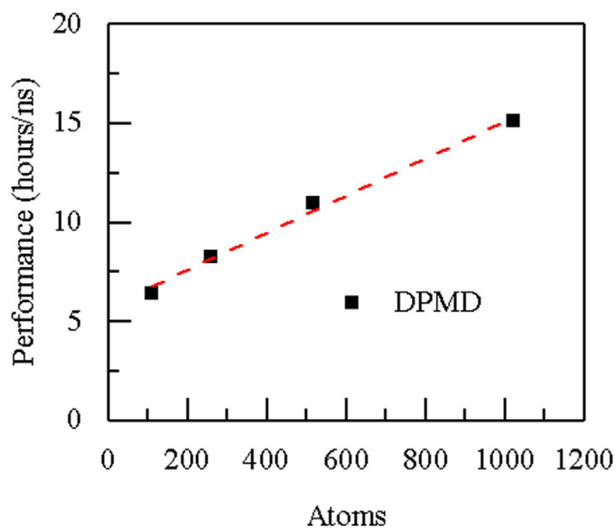

Figure 8. The computational cost of DPMD simulations as a function of system size. All calculations are performed with Intel Xeon CPU E5-2650 and Nvidia Tesla GPU P100.

4. Conclusions

Molecular simulations were conducted to investigate the local structure and basic properties of molten magnesium chloride using ML potential. The ML potential was trained using datasets extracted from FPMD calculations. The ML potential well reproduced the DFT energies and forces in both the training dataset and validation data set with the RMSEs less than 2×10^{-3} eV/atom for the energies and 5×10^{-2} eV Å^{-1} for the forces. Moreover, the local structural and basic properties of molten $MgCl_2$ are investigated between 1000 and 1400 K using the ML potential. Predicted structure information of DPMD is similar to previous FPMD results. The simulated density and self-diffusion coefficient are in good agreement with experimental or FPMD values.

Compared with FPMD simulations, the benefit of performing MD simulations with ML potential is that the computational efficiency is significantly improved while the accuracy is kept. In addition, the transferability of the ML potential has been shown by simulations at multiple temperatures. It means that MD simulations can be conducted using a larger size of simulation cell, a longer period of simulation time, and a smaller interval of temperature with accuracy comparable to that of FPMD and with efficiency similar to that of empirical potentials.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

deep potentials, machine learning, magnesium chloride, molecular dynamics simulations

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